

September 4, 2026

Dr. Chun-Wei Yang
Handling Editor
Scientific Reports

Dear Dr. Yang,

Please find the focused revision of *RankCloak Conceals the Surface Form of Synthetic Cryptographic Artifacts in Language Model Generated Text* (submission 0170c7b2-1065-4344-a5d5-b839b51ac22c). Thank you and Reviewer 2 for the constructive second assessment. We retained the first-round revisions and completed the additional experiments requested in this round.

RankCloak addresses an artifact-specific problem: transforming recognizable synthetic cryptographic artifacts into context-dependent token choices so their hexadecimal, UUID, base64, nonce, tag, ciphertext-like, or signature surface form is not directly exposed in generated cover text, while an authorized receiver reconstructs the artifact under a shared protocol configuration. Steganalytic classification is a distinct endpoint. The new results show that current detectors can often identify encoded covers, but such detection does not itself reveal or recover the concealed artifact.

The revision adds strict pre-feature deduplication with zero leakage under the declared audit, 250 human-authored secondary controls, genuine leave-one-model-family-out tests, data-adaptive text detectors, an exact-model-aware attacker, and validation-frozen low-FPR analysis with exact false-positive counts. It also adds entropy-gated embedding and a matched comparison of approximately 4-bit and 8-bit quantized versions of the same pinned Qwen model, including recovery and cross-quantization detection. The complete declared computational matrices and their validation finished successfully without generation execution failures. Adverse outcomes remain visible: human-control false positives reveal domain sensitivity, strict entropy gating lowers capacity and lengthens covers while leaving six payloads incomplete, model-aware detection remains strong, visible-text transport is weaker than token-preserving recovery, and the paired analysis does not support an 8-bit recovery advantage.

The main manuscript retains its artifact-hiding identity, equations, algorithms, primary results, and principal figures, with the new evidence summarized in one compact Results subsection and table. Condition-level methods, thresholds, counts, calibration, provenance, and limitations are provided in the Supplementary Information. The paper does not claim a cryptographic secrecy proof, universal undetectability, or robustness under every text transformation.

The existing GitHub and Zenodo statements remain truthful. Final V3 Zenodo synchronization, archived-version confirmation, and DOI wording synchronization are listed as manual pre-submission steps and are not represented as already complete.

Thank you for considering this revision.

Sincerely,

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